

Miguel Shinyenyi

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PROFESSIONAL SUMMARY

Backend Software Engineer with 3+ years of experience building scalable, data-intensive systems in Java and Spring Boot. Proven track record delivering multi-tenant ERP platforms, RESTful microservices, and automated data pipelines processing hundreds of daily transactions across distributed systems. Strong foundation in event-driven architecture, PostgreSQL optimisation, TDD, and secure access control. Writes technical deep-dives on the systems he builds and maintains an open-source fintech settlement engine as a working example of the same practices.

WORK EXPERIENCE

Software Engineer — Global Link+

Sep 2023 – Present
Utah, USA (Remote)

Multi-tenant ERP logistics platform serving 10 B2B clients, with integrations across Shopify, Amazon, and Walmart.

- Designed and maintained RESTful backend services in Java Spring Boot powering a multi-tenant ERP platform, enforcing strict data isolation across 10 B2B clients and their customer bases, processing hundreds of orders daily.
- Built and optimised backend APIs for a real-time admin dashboard tracking client shipment volumes, order activity, and logistics KPIs, combining live PostgreSQL queries with pre-aggregated data for historical reporting.
- Integrated Stripe for secure payment processing, improving transaction reliability and the client-facing payment experience.
- Implemented scheduled background jobs to automate client report generation, aggregating daily order and shipment data across 10 clients into structured summaries and eliminating manual reporting overhead.
- Introduced Redis caching for frequently-accessed, low-volatility dashboard data and optimised PostgreSQL queries through indexing and query planning, reducing API latency by 40% and cutting redundant database load under peak concurrent usage.
- Architected and delivered a secure file management system storing tens of thousands of documents (contracts, product images, shipping records) with role-based access control enforced at the API layer.
- Designed a multi-tenant PostgreSQL data model with tenant-scoped schemas and query filtering, ensuring data integrity and isolation across all client tiers without relying on application-layer safeguards alone.
- Contributed to CI/CD pipeline setup by writing Maven build scripts to automate compilation, testing, and packaging for consistent, repeatable deployments.
- Improved system observability by integrating JVM metrics and backend health checks into monitoring dashboards, surfacing performance bottlenecks before they reached clients.
- Participated in code reviews and pair programming, upholding clean-code principles and consistent engineering standards across the Java codebase.
- Collaborated with product and operations teams to translate logistics workflows into backend features, bridging business requirements and technical implementation in a cross-functional startup environment.

Software Engineer — IBM Research

Jun 2023 – Aug 2023
Nairobi, Kenya

Backend services supporting real-time molecular visualisation and AI-driven computation for scientific research workflows.

- Developed Flask-based APIs for real-time molecular visualisation and AI-driven computation, working directly with researchers to diagnose integration issues and improve data accuracy and latency.
- Built and tuned PostgreSQL queries for analytical workloads, applying indexing and query planning to support high-read scientific data pipelines.
- Containerised backend services using Docker, ensuring consistent deployment and environment reproducibility across research and production systems.

Software Engineer — Quikk API

Sep 2022 – May 2023
Nairobi, Kenya

Spring Boot microservices powering an analytics-driven API platform.

- Built and optimised Spring Boot microservices with JWT authentication, keyset pagination, and database indexing for secure, performant API access.
- Integrated PostHog for real-time user telemetry and performance metrics, supporting proactive troubleshooting and data-driven product decisions.
- Conducted unit and integration testing following TDD principles, maintaining high test coverage and contributing to service reliability.

PROJECTS & WRITING

Invoice-Financing — open-source settlement & reconciliation engine

github.com/Miguel-Shinyenyi

- Personal fintech backend project: an idempotent settlement engine with a transactional outbox, double-entry ledger consistency checks, and concurrency-safe idempotency-key handling under load.

Technical writing — personal site, dev.to, Hashnode

miguel-shinyenyi.github.io/miguel-site

- Writes and publishes deep-dives on backend systems design, drawn from real production and portfolio code, cross-posted to dev.to and Hashnode.

TECHNICAL SKILLS

Languages & Frameworks Java, Spring Boot, JavaScript, TypeScript, Fastify, Flask

Backend & Architecture RESTful APIs, microservices, event-driven architecture, pub/sub patterns, synchronous & asynchronous flows, multi-tenancy, stateful vs. stateless systems

Databases & Caching PostgreSQL, MySQL, MongoDB, DynamoDB, Redis — indexing, query optimisation, keyset pagination, caching strategies

Messaging & Streaming Familiar with Kafka and RabbitMQ event-driven patterns; publish/subscribe, streaming pipelines, message queuing

Testing Unit testing, integration testing, TDD, test coverage, regression testing

DevOps & Tooling Docker, Maven, Git, CI/CD pipelines, build scripting, AWS, Linux

Observability JVM metrics, health checks, PostHog analytics, log-based debugging, performance monitoring, incident triage

Networking Fundamentals OSI model, reverse proxies, load balancing, API routing, connection multiplexing, caching layers

EDUCATION

Bachelor of Science, Computer Science

Gretsa University

Thika, Kenya